



KIRLOSKAR CHILLERS PRIVATE LIMITED

Enriching Lives

ENGINEERING BULLETIN

From :	SNM	Engg Bulletin No. :	EB/13-14/10
Approved By:	SSN	Date of Release :	17 th Feb, 2014
Subject :	Information bulletin – Various Operating Modes of WC Chillers		

This is the information bulletin which highlights difference in specification / performance of Brine & Dual mode chillers than the normal one wherever applicable.

Basically KCPL offers water cooled chillers for following various operating modes,

1. Normal mode.

Chillers for which Evap. Leaving Temp is within 4.5 to 10 deg C (max. can go up to 15 deg C in special cases) and Cond. Entering Temp is within 18 to 38 deg C.

Performance & specification of these chillers will be as per released selection software, GAs, Data sheets etc.

2. Brine only mode.

Chillers for which Evap. Leaving Temp is within -20 to 4 deg C and Cond. Entering Temp is within 18 to 38 deg C.

These chillers are only meant for brine applications only.

Performance of these chillers (for brine conditions) should be referred from selection software & released Rating chart (EB/13-14/08)

*** Performance of these chillers at normal temperatures SHOULD NOT be referred from selection software***

Internal testing & customer witness testing (if at all required) will be done at Evap. Leaving Temp 7 deg C and specified Cond. Entering Temp. Performance at these conditions SHOULD BE committed / informed to customer only after consultation from Engineering Dept.



Water side Pressure drops for Heat exchangers SHOULD BE taken from Engineering Dept.

Rarely chillers may need to run at two different brine temperatures (e.g. -5 deg C & -10 deg C), in such cases also it will be referred as brine only mode, not as dual mode.

For overall dimensions & weight refer standard Physical data of corresponding model. Note that these may or may not vary during detail engineering which will be done after receipt of order. But definitely new dimensions and weight will be equal to or less than the standard model.

*If Evap. Leaving Temp is below -10 deg C, material provided for Evaporator shall be as per ASME STD. (BQ Material),
-Plate material as per SA 516 Gr.70
-Pipe Material as per SA 106 Gr. B.*

If Evap. Leaving Temp is below -10 deg C, Insulation thickness provided for Evaporator shell shall be 51 mm.

3. Dual Brine mode.

These chillers will work at both, Normal mode (above pt. no.1) as well as Brine mode (pt. no. 2)

*** These chillers can only be offered up to brine Evap. Leaving Temp of -10 deg C. That means brine mode of dual mode chillers will only be possible minimum up to -10 deg C ***

Performance of these chillers at both modes should be referred from selection software.

Internal testing & customer witness testing (if at all required) will be done at Normal mode only.

For overall dimensions & weight refer standard Physical data of corresponding model.



4. Heat Pump mode.

Chillers for which Evap. Leaving Temp is within 4.5 to 10 deg C (max. can go up to 15 deg C in special cases) and Cond. Entering Temp is within 40 to 60 deg C.

These chillers are only meant for heat pump applications.

Performance of these chillers (for heat pump conditions) should be referred from Rating Chart. Rating Chart will be released very soon.

Internal testing & customer witness testing (if at all required) will be done at Evap. Leaving Temp 7 deg C and Cond. Entering Temp 32 deg C. Performance at these conditions SHOULD BE committed / informed to customer only after consultation from Engineering Dept.

Condenser shall be designed for higher pressure & temperature. And thickness of condenser subparts will be higher than std. chiller. Also safety valve set pressure will be higher.

For overall dimensions & weight refer standard Physical data of corresponding model. Note that these may or may not vary during detail engineering which will be done after receipt of order. But definitely new dimensions and weight will be equal to or less than the standard model.

5. Dual Heat Pump mode.

These chillers will work at both, Normal mode (above pt. no.1) as well as heat pump mode (pt. no. 5)

Performance of these chillers (for heat pump conditions) should be referred from Rating Chart. And for Performance at normal mode should be referred from selection software.

Internal testing & customer witness testing (if at all required) will be done at Normal mode only.

Condenser shall be designed for higher pressure & temperature. And thickness of condenser subparts will be higher than std. chiller. Also safety valve set pressure will be higher.

For overall dimensions & weight refer standard Physical data of corresponding model.